

Scalability & Future Plan

Date	30 SEPTEMBER 2026
Team ID	6ab3ecd3a238dc999eb3fcc2
Project Name	ComicCraft AI
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	External AI/provider dependency	Generation can fail on quota, rate-limit or provider availability errors.	High
2	Local generated-file storage	Generated assets/results are stored in the project filesystem.	Medium
3	Single-process generation path	Five panel images are generated in sequence in the current pipeline.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local/single-app deployment	Add multiple workers and a background job queue for generation.
2	Data Storage	Filesystem + JSON result files	Move assets to object storage and metadata to a database.
3	Performance	Sequential five-panel generation	Use queued/concurrent jobs, caching and provider scaling.
4	Security	API keys loaded from .env	Use managed secrets, authentication and rate limiting in production.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Persistent storage + background job queue	To be planned	Support larger workloads and more reliable jobs.
Phase 3	Authentication, multi-user support and cloud storage	To be planned	Enable broader and safer deployment.
Phase 4	Provider/model scaling, monitoring and analytics	To be planned	Improve reliability, performance and maintainability.